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Toward a Recipe for Deep versus Surface Level Tailoring: Mixed-Methods Validation of Message Features to Reduce Sugary Beverage Consumption

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Abstract

Targeted marketing contributes to the overconsumption of sugary beverages, which contributes to obesity and diabetes disparities among African American and Latino populations in the U.S. Health communicators can similarly use culturally-tailored messages to decrease sugary beverage consumption among these groups, yet the specific strategies to operationalize cultural tailoring—the message components essential for such tailoring—are ill-described. We sought to identify and validate authentically-created, culturally tailored messages using a multiple-step mixed-methods approach. First, we used a snowball approach to identify nutrition education messages targeting ethnic minorities about reducing sugary beverage consumption (N=85). Via content analysis, we assessed message features (character gender and race/ethnicity), level of change of the appeal (individual or social), and level of cultural tailoring (surface level tailoring in the form of matching character gender and race/ethnicity with target audiences versus deep structural tailoring in the form of appealing to values is an effective message strategy). The highest-rated videos were then validated by a sample of the target audience using a quantitative survey and qualitative comments (N=76). Results inform theorizing on message tailoring and provide a validated pool of culturally relevant messages intended both to reduce intentions to consume sugary beverages and to engage in social change actions.

1. Introduction

Overconsumption of sugary beverages contributes to obesity and diabetes disparities experienced by Black and Latino youth and young adults, evidenced, for example, by the disproportionately high rates of sugary beverage consumption among African Americans and Latino groups from toddlerhood on, relative to white populations (Bleich & Vercammen, 2018; Freeman et al., 2016; Rosinger et al., 2017a, 2017b). Sugary beverage consumption is heavily influenced by context and the social and commercial determinants of health (Barquera & Rivera, 2020; Khandelwal & Salazar, 2020; Pedroza-Tobias et al., 2021). Beverage companies use targeted marketing to influence the beverage choices of vulnerable populations, including the high availability and low price of sugary drinks in low-income

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neighborhoods with high proportions of ethnic minority residents and placement of sugary drinks on the shelves close to youth's eye level and in checkout lines (Grier & Kumanyika, 2010). Moreover, the sugary beverage industry deliberately targets youth of color using a variety of advertising strategies including: tailored messages that appeal to ethnic identity and cultural values; partnerships with African American and Latino sports and music celebrities to build brand awareness; and placing advertisements in Spanish-language media (Backholer et al., 2021; Barnhill et al., 2022; Zhou et al., 2020). We argue that the targeted marketing must be both an essential target of health communication, and that health communicators can learn from the corporate message strategies to design effective nutrition messages.

1.1 Message Design for Multilevel Change

Prominent public health frameworks acknowledge the multi-level nature of health determinants and the implications of public health interventions at different levels (Frieden, 2010; Golden & Earp, 2012), and there is substantial evidence that corporate, social, and environmental factors affect sugary beverage consumption (Chaffee et al., 2021). Nonetheless, approaches to communicating about dietary health and specifically about consuming sugar generally rely on appeals to individual behavior change. Individual level appeals can affect individuals' awareness, knowledge, attitudes, self-efficacy, and intentions to change behavior, but such messaging may not be as effective in the presence and influence of structural and environmental constraints on behavior (Fishbein & Cappella, 2006). Effective communication strategies thus must include appeals to social change; this would be consistent with an ecological approach to communication (Moran et al., 2016). Such appeals may include calls for civic engagement, political action, or consumer engagement. Yet even as the constraints of individual behavior change messaging are recognized, social change has not been a primary focus of health communication messages. For example, a systematic review of 157 health promotion interventions found that 95% of the reviewed studies described individual-level activities; not surprisingly, then, only 20% of studies reported community-level behavior changes (Golden & Earp, 2012). Specifically, in the context of dietary health and health disparities, the role of the food and beverage industry (i.e., targeted advertising of junk foods, public relations efforts, product and store placement, and other marketing and policy strategies) has been increasingly recognized as the "corporate and commercial determinants of health" (Maani et al., 2020). Corporate and commercial factors as such have not been paid enough attention in research using the public health framing of "social determinants of health", which underestimate corporate responsibility for health.

Some intervention research has sought to support vulnerable populations in understanding the very compelling corporate messaging that blanket targeted communities. For example, media literacy, measured by an individual's ability to analyze, process, evaluate, and criticize media messages, has recently attracted significant attention and resources from health educators and researchers. In the context of health, media literacy interventions have been demonstrated effective in improving youth's critical thinking to reduce negative impacts of food marketing on their food requests (Austin et al., 2018) as well as their consumption of fruits and vegetables (Austin et al., 2020). Media literacy training specifically focused on sugary beverage marketing has enabled adults to be more skeptical toward the targeted marketing and to consume less sugary beverages (Chen et al., 2020).

However, simply raising awareness of targeted marketing may not be enough to activate engagement in the absence of specific recommended actions or support—or in the presence of conflicting information or message features. For example, a recent cross-sectional study examined the association between sugary beverage media literacy and beverage consumption found that Black and Latino youth are more trusting of food and beverage advertisements than their white counterparts (Roesler et al., 2021). Even though youth of color clearly acknowledged the targeted nature of advertising for unhealthy foods and beverages, demonstrated in a recent qualitative study, they considered the products healthy because the endorsement of prominent athletes created a healthful association (Miller et al., 2021). Another recent study shows that low-income parents of young children are motivated to support sugary beverage reduction policies by messages that acknowledge the need for higher-level support for families and communities to engage in healthy behaviors (Cannon et al., 2021). Therefore, ecological approaches to health communication must center the socioecological (including corporate) determinants of health and deflect blame from individuals to external actors, while empowering individuals to advocate for changes in the systems and environments (Moran et al., 2016). Dresler-Hawke and colleagues emphasized that individual efforts to adopt healthy dietary behaviors are more likely to work in a supportive environment when the process is shifted in the sentiments of the whole community. Thus, they suggested that communication strategies need to be consistent across all societal levels to make meaninful behavior changes (Dresler-Hawke & Veer, 2006). A recently published systematic review also called for communication efforts to build a social change movement to reduce sugary beverage health risks for Americans (Kraak et al., 2022). The inclusion of explicit appeals to social change, with specific recommended actions, is thus critical for effective messaging.

1.2 Culturally-Relevant Message Tailoring

As we consider the design of effective messages to counteract corporate influence on sugary beverage behaviors, it essential to consider the role of cultural tailoring. Culturally-tailored messages have been effectively deployed by the food and beverage industry as part of an overall strategy to target African and Latinos and encourage unhealthy consumption (i.e., of sugary beverages) (Backholer et al., 2021). But, messages that integrate identifiable cultural cues also have been more effective than standard messages in shaping healthy behaviors (Hawkins et al., 2008; Noar et al., 2007). Cultural tailoring has been shown to be effective in the context of diet (Bryan et al., 2016; Nierkens et al., 2013), smoking cessation (Nierkens et al., 2013), physical activity (Hawthorne et al., 2008; Nierkens et al., 2013; Renzaho et al., 2010) and in cancer communication in general (Huang & Shen, 2016).

Research has distinguished between two levels of tailoring to achieve cultural appropriateness: *Surface level cultural tailoring* approaches are those in which attempts are made to match message features with basic visual factors about the target population, such as language, ethnicity, gender, and clothing (Ken Resnicow et al., 2009), while *deep (structural) level cultural tailoring* incorporates the values, beliefs, or norms shared by a group (Armstrong et al., 2005; Kreuter et al., 2003; K Resnicow et al., 1999). Theoretically, surface tailoring is hypothesized within the Elaboration Likelihood Model (ELM) to operate as peripheral cues to processing that influence participants with low involvement; in contrast, deep tailoring triggers central processing that generates greater attention from highly involved participants, and thus produces more powerful persuasion effects (Huang & Shen, 2016; Jensen et al., 2012). Yet the evidence base actually examining the relative contributions and persuasive mechanisms of surface versus deep level cultural tailoring is weak: More meta-analyses or systematic reviews than individual studies have explicitly distinguished between surface and deep tailoring, and consensus in results have not been achieved. For example, Huang & Shen 2016 examined 36 articles within 58 experimental pairs concluded that deep tailoring had a significantly stronger effect than surface tailoring in promoting cancer screens and reducing cancer disparities among diverse populations (Huang & Shen, 2016). An additional systematic review supported this results in interventions for chronic disease among Asians in 21 articles across 16 unique interventions (Heo & Braun, 2014). However, a separate clinical trial that tested the functions of surface and deep tailoring in enhancing recall of oral health information among Mexican American did not identify significant difference between the two levels of tailoring, though both levels were effective (Singelis et al., 2018). Moreover, because the message tailoring in meta-analyses or systematic reviews were manually coded by researchers for their included studies, the results may reflect biases. For example, Huang and Shen 2016 coded studies that employed both surface and deep tailoring as deep tailoring studies, confounding the effects of surface and deep level message features (Huang & Shen, 2016; Singelis et al., 2018). Another study that incorporated cultural values to adapt a healthy diet text message for Latino adults similarly concluded that the deep level tailoring could be effective at motivating dietary change (Cameron et al., 2017). However, that study conflated surface and deep structural tailoring since Latino-targeted messages were written in Spanish only. Thus, part of the challenge of testing the relative contribution of surface and deep level structural tailoring may be effectively identifying or designing research stimuli that capture only surface and only deep level structural factors. The present study aims to address these challenges by explicitly identifying the features that correspond to deep versus surface level tailoring and creating a pool of validated messages that can be used for future message testing.

1.3 Integrating tailoring and multilevel message appeals

Another limitation of most existing message tailoring research is a singular focus on individual level behavioral outcomes. Consistent with the reasoned action approach (Fishbein & Cappella, 2006), message design research has sought to match message features with individual preferences, or in the context of group level tailoring, preconceived group-level preferences, but with the goal of changing individual behaviors or behavioral precursors (Tan & Cho, 2019). For example, Bryan and colleagues first framed healthy eating as consistent with adolescents' values (i.e., the desire for autonomy from adult control and the pursuit of social justice) and then reframed junk food and food marketing that targeted to youth or poor as inconsistent with these values. They found values-based health communication messages can create low-cost, sustained behavior change on a population scale (Bryan et al., 2016, 2019). While the authors of that study did not frame the messages in terms of tailoring, we argue that values-based messaging is an essential component of deep structural tailoring, especially in the context of culturally appropriate message tailoring. However, research is limited in exploring if deep structural tailoring—specifically, appealing to group values—may be effective for changing not just individual behavioral intentions and actual behaviors, but also to increase awareness of targeted marketing as a social justice issue and a sense of empowerment to engage in social change.

1.4 Study Objective

Given the strong body of evidence for group-level tailoring effects on individual behaviors, and calls for evidence for their effects on collective action, there is a need to better understand and disambiguate the mechanisms for surface and deep level cultural tailoring message effects. Essential to that research is a better understanding of how to manipulate message features to engender surface versus deep level tailoring. While it is possible for researchers to create original content for message testing purposes, the results lack external validity, especially when considering messages targeting ethnic minority and low-income populations who are underrepresented in the academe, and as such have limited practical utility (Ramírez, 2013). Thus, we sought to identify real-world messages that aim to reduce sugary beverage consumption and to validate the operationalization of specific message features, as perceived by the audience. Our health focus is on diet-related disparities experienced by Black and Latino populations in the U.S., owing much to their excess consumption of sugary beverages (Bleich & Vercammen, 2018; Freeman et al., 2016; Rosinger et al., 2017a, 2017b). Specifically, we sought to identify healthy diet messages that target African American and/or Latinos. Our overall goal for this study was to build a pool of messages that could be used in future studies to test specific hypotheses about the relative efficacy of these message features to reduce sugary beverage consumption and increase civic engagement among youth and young adults of color.

2. Methods and Results

The present study had two distinct phases. First, we sought to identify real-world messages that targeted African American and/or Latinos with messages about the harms of sugary beverage consumption. We used a snowball approach to identify candidate messages, and then conducted content analysis to examine key message features. In the second phase, we validated the key features with members of the target audience who are low-income young adults of diverse race/ethnicity using a combination of closed-ended survey questions, open-ended questionnaires, and guided discussions. This study was approved by the institutional review board (IRB) of University of California, Merced with the reference number UCM15-0017. Since the identification and validation processes were iterative, we present the methods and results sections for these two processes together.

2.1 Phase I: Identification and Content Analysis

2.1.1 Phase I-a: Identification of culturally-tailored nutrition education messages aims to reduce sugary beverage consumption—Together, African Americans and Latinos comprise the largest ethnic minority group in the U.S., suffer disproportionately from preventable diet-related disease, and most ethnically targeted messages have focused on these groups (United State Cencus Burean, 2021; Williams et al., 2012). Therefore, we identified potential messages via generic internet search engine (Google) and YouTube using different combinations of the keywords “sugary drink” or “sugary beverage” or “sugar sweetened beverage” and “Latino” or “Latina” or “Hispanic” or “African Americans” or “ethnically targeted” and “overweight” or “obesity” or “diabetes” and “American” or “United States”. We also conducted targeted searches on websites we knew to have nutrition related messaging, such as *California’s Obesity Prevention Project* and the *U.S. Centers for Disease Control and Prevention*, as well as those specifically targeting ethnic minorities, such as *Salud America!*. We then used a snowball approach to identify other relevant resources from the search results; we looked for messages produced by the same organizations or sponsored by similar organizations as well as reference lists on the websites where the search result messages were embedded. Our snowball search also led us to social media sites of the sponsoring organizations, where we looked at the pages of the people who had interacted with the posted messages to identify other potentially relevant resources.

For the initial message identification phase, we had the following inclusion criteria: (1) health education messages about sugary beverage consumption targeting African Americans or Latinos; (2) from reasonably credible resources (e.g., national/subnational governments, non-profit organizations, health care systems, or higher education institutions); (3) because prior studies have established that acculturation is associated with increased risk of poor diet (Ard et al., 2005; Ramirez et al., 2021), we specifically focus on more-acculturated African Americans and Latinos. Consistent with prior studies (Park et al., 2016; Ramírez et al., 2018), we consider English-language ability as a rough proxy indicator of behavioral acculturation to “American” culture, and thus at a higher risk of unhealthy diet. Therefore, we focused our search for messages presented in English; (4) incorporate audio or visual cues to culture (“surface level tailoring”; e.g., visibly Black/Latino characters; for Latino-targeted messages, Spanish-language accents or language cues) or appeals to values (“deep structural tailoring”; e.g., familism, social justice, standing up to exploitative authority (Cuellar et al., 1995; Knight et al., 2010; Schwartz, 2007). Finally, previous research found that cultural tailoring effects are larger when the message is delivered through audio or video compared to print media (Huang & Shen, 2016). Therefore, we only searched for videos.

2.1.2 Phase I-a: Results—We identified 85 unique videos from 9 unique campaigns; the videos ranged in length from 30 seconds to 9 minutes. Additional searches resulted in the same videos under different YouTube channels or in low-quality amateur productions, indicating that we had reached saturation (Neuendorf, 2016; Saunders et al., 2018). We then applied the following exclusion criteria: (1) Too short (<30 seconds), as these videos could not include sufficient content for our purposes; (2) The message appeal was too graphic and consequently invoked disgust or similar negative emotions, as these emotions are known to cause particular types of reactance that impede message processing (Leshner et al., 2009; Peters et al., 2013); and (3) production quality was poor. After exclusions, we retained a total of 19 videos ranging in duration from 1 minute to 5 minutes and 50 seconds (16 from *The Bigger Picture Campaign*, and 3 from Kaiser Permanente’s *Rethink Your Drink Campaign*, Santa Clara, CA) for the next stage.

2.1.3 Phase I-b: Content analysis of key message features—
We conducted systematic content analysis of key message features of the 19 videos; specifically coding for: (1) character gender and race/ethnicity, (2) video duration, (3) the format of videos (narrative/story telling or non-narrative/didactic), (4) level of cultural tailoring (deep or surface, or both), and (5) level of change in the message appeal (individual or social, or both). One researcher (DC) coded all the videos. The other two researchers (ASR and MZ) then reviewed the coding; differences in opinion regarding the codes were identified and resolved by consensus. The objective of the coding was both theoretical and practical. Theoretical, in that our primary goal for this study was to characterize the features that engender cultural tailoring. Practical, in that we sought to build a pool of exemplar messages that could be used for future experimental testing. Messages needed to be appropriately equivalent in terms of duration and other essential message features. As such, following the systematic content analysis of the 19 videos, we excluded videos that did not conform to the theoretical framework (e.g., did not include clearly identifiable surface or deep tailoring features). We further excluded videos that did not include explicit calls to action at the individual or social levels. Finally, because previous research has found that narrative messages are more effective in culturally tailored health communication (Huang & Shen, 2016), and we sought to build a pool of relatively videos that varied only on the key theorized variables (gender and ethnicity of the characters, other surface-level features, key values that represent deep structural tailoring, and individual/social level appeals), we dropped the non-narrative videos.

2.1.4 Phase I-b: Results—
The final selected videos (N=8) had the clearest articulation of values (deep structural tailoring) and execution (surface level tailoring), as determined by overall agreement, and also by the relative uniformity of production style, format, and duration. All 8 videos come from *The Bigger Picture Project*, a diabetes prevention campaign launched by UCSF's Center for Vulnerable Populations in collaboration with Youth Speaks (Rogers et al., 2014). *The Bigger Picture Project* is a collection of spoken word poems and videos created by racial and ethnic minority youth; the texts and accompanying images highlight the manipulative practices of the food and beverage industry and invite youth to defy their authority by consuming less sugar and by participating in advocacy efforts encouraging their community members to consume less sugar. These messages were created *by* and *for* ethnic minority populations. Although this was not an explicit criterion during the message identification phase, the inclusion of members from the target audience in the process of conceptualizing and designing the messages adds ecological validity for message design (Guo et al., 2020).

2.2 Phase II: Target Audience Validation

2.2.1 Validation process and data analyses—
The objective of the second phase was to validate researchers' interpretations of the messages with members of the target audience. Participants (N=76) were enrolled to test the eight selected videos. Each participant was randomly assigned to view two videos on their phone or laptop. Participants watched the videos online and responded to a set of questions on an online survey platform as part of an in-class research activity in an undergraduate public health course. Specifically, we assessed: (1) Participants' gender (to examine the extent to which matching gender with

the message character influences perceptions of the message); (2) Participant's ethnicity (as with gender, to examine matching); and (3) Participants' perceptions of the gender and race and ethnicity of characters in the message. We further assessed the (4) main message of the video and (5) feelings evoked by the video, using open-ended prompts. We specifically asked if (6) the message mentioned family; if yes, then we asked participants (7) to describe the ways in which family was mentioned. Two investigators (MZ and DC independently coded the open-ended questions, and the disagreements were solved with the third investigator (ASR). For the deep-level tailoring, we coded the identification of familism as "1" if the participants thought the videos mentioned familism in question (6) and (7). We coded the identification of values of social justice/stand up against authorities as "1" if the participants mentioned the unfairness for the sufferings of the families in the video or expressed intention to change the status quo in question (5). For the level of appeal in ecological framework, we coded the identification of individual behavior change as "1" if participants thought the characters should change their diet habit to main good health and the identification of social determinants of health as "1" if the participants mentioned that the food choices were influenced by the social factors in question (4). After all the coding, we compared the identified message features between researchers and target audiences, if more than 54% of participants have the same coding as researchers, we considered as consistency (Pew Research Center, 2018).

Because the selected video were in narrative format, in addition to assessing participants' perceptions of the cultural tailoring embedded in the messages, we assessed key indicators of narrative persuasion based on the Extended Elaboration Likelihood Model (EELM) (Murphy et al., 2015; Slater & Rouner, 2002), as follows: (8) Identification with the character in the video, measured using a 5-point, 6-item Likert scale with responses ranging from Strongly Agree=5 to Strongly Disagree=1 (Cohen, 2001; Phua, 2016); (9) Engagement with video, measured using a 5-point, 5-item Likert scale (Kim, 2019); and (10) Perceived similarity with main character in the video, measured using a 5-point, 3-item Likert scale (Cohen et al., 2018). We present summary descriptive statistics for these quantitative measures. Though this current study is not hypothesis-based, we did some preliminary statistical testing to identify the extent to which perceived persuasion varied across messages that matched and did not match participants' gender and racial/ethnic identification. We conducted two separate two-way ANOVA for this purpose. The overall narrative persuasiveness (average score of the Identification, Engagement, and Similarity) was the dependent variable. The gender (i.e., male and female) of participants and the gender of characters in the video were the two independent variables in the first ANOVA. The race/ethnicity (Latino and non-Latino) of participants and race/ethnicity of characters in the video were the two independent variables for the second ANOVA. Analyses were completed using Stata 16 (StataCorp, TX).

2.2.2 Phase II results—Most of the participants in the video validation study identified as women (N=55/76, 72%) and as Latino, Hispanic, or Mexican American (N=53/76, 70%). The two-way ANOVA results for the overall narrative persuasiveness based on the match of participants' and characters' gender and race/ethnicity are presented in Table 1. We did not find any interaction effects. That is, videos with male characters received significantly higher score than video with female characters (p = 0.019) regardless the gender of participants and videos with Latino characters received higher score than video with non-Latino characters regardless of participants' race/ethnicity.

There was broad agreement between researchers and the target audience on most of the open-ended responses (Table 2). Most surface level tailoring features including identification of the characters' gender and race/ethnicity had high agreement. The exception was identification of the main character's ethnicity in *Bottled Up*: only 44% participants identified the main character as Latina. With respect to the values-based appeals—the operationalization of deep structural cultural tailoring—there was mixed agreement. Researchers and the target audience agreed on messages that expressed the value of social justice, but there were discrepancies in perceptions of the values of familism: Three of the videos (*Monster*, *The Corner*, and *Empty Plate*) were perceived by members of the target audience as expressing the value of "familism," whereas they were not perceived as such by the researchers.

For the message of individual appeals, we found relatively good consistency, since for each video that coded with "No" by researchers, fewer than 54% of participants believed it was focused on individual behavior change, and for those coded with "Yes" by researchers, more than 54% believed were individual appeals, except the video *The Corner* (50%). Finally, researchers and participants consistently identified the social determinants of health as a key appeal in all messages, except in *Hear No*. Table 2 also shows the overall score of narrative persuasion for each video. The video *Empty Plate* was top ranked, followed by *Lost in Translation* and *The Taste of Home*. *Monster* received the lowest score.

Based on the validation results, we dropped two videos from the final pool. *Monster* was eliminated because 1) the overall narrative persuasive score was very low; and 2) though there was a relative agreement between researchers and target audience in terms of the race/ethnicity of the characters, neither of the two girls was clearly identifiable as Latino or White, which made the target population of this video ambiguous. Furthermore, a few participants described the atmosphere of this video as creepy (*"It made me feel scared and intimidated. Throughout the whole video I felt emotions such as fear, anxiety, and discomfort. This was caused because of the strength and tone of their voice. The imagery and the color grading of the video was very dark."*) or ineffective (*The video was a little bit hard to watch because it was cringe. It was funny at times because of the dad's acting. The heart felt poem didn't help at all. After watching the video, if I drank energy on a daily basis, I would've keep drinking and not stop."*). We dropped *Empty Plate* even though it received highest narrative persuasive scores because there were concerns about confounding of the target constructs: The focus was primarily on barriers to healthful eating for residents of an agricultural region, with soda consumption garnering only a passing mention. Moreover, both *Monster* and *Empty Plate* had substantial variability in the target audience's perception of familism. Finally, since the social factors were only mentioned toward the very end of the messages in an ambiguous way in the video *Hear No,* we adopted participants' interpretation for this message as an individual only stimulus. We kept the video *Bottled Up* because of the relative consistency in perceived values between researchers and participants, despite the disagreement on character's ethnicity. We kept the video *The*

Corner because of the consistency on the surface-level characters between researchers and participants and the relative high score of the narrative persuasion, despite the disagreement on familism and individual behavior change.

3. Discussion

The quality of stimuli is the bedrock of human behavior research. As African Americans and Latinos experience disparities in health, there is an urgent need to understand the factors that influence their receptivity to health promotion messages. Much research in health communication has established the importance of tailoring messages to the target audience, and in the context of cultural tailoring, theorizing has pointed out deep versus surface level tailoring (Huang & Shen, 2016). Yet there remains a dearth of evidence of the relative efficacy of deep or surface tailoring. We propose that one of the reasons for this lack of evidence has been the lack of clarity about how such message features should be operationalized. In this study, we used a multi-step, mix-methods approach to systematically identify, classify, and validate a set of real-world nutrition education messages that can be used in future studies to test these message features. Our overall goal for this study was to build a pool of messages that could be used in future studies to test specific hypotheses about the relative efficacy of these message features to reduce sugary beverage consumption and increase civic engagement among youth and young adults of color.

Although the present study was not designed to test these hypotheses, but rather to validate key message features, we did conduct some preliminary message effectiveness testing to triangulate findings across methods. Participants did not get significantly higher score when viewing videos of their same gender or race/ethnicity suggests that surface-level tailoring focusing on a single demographic characteristic might not be sufficiently tailored to engender effects. The limited number of data we collected does not allow for meaningful four-way ANOVA analyses that incorporate both gender and race/ethnicity of participants and video characters in one analysis to further investigate the interaction between gender and race/ethnicity. However, a separate descriptive analysis that included all variables shows the highest overall persuasion score among Latinas when they viewed female and male African American video characters. This result further suggests the undependability of surface only tailoring. Our message pool provides resources for future studies to compare messages tailored by surface only, deep only, or the combination of the two, whether used as exemplars to create unique messages or to use as-is.

Based on the validation results, we dropped the video *Monster*, mainly because the low score of narrative persuasion it received and the greatest discrepancies regarding the identification of familism between researchers and participants. The iterative process of discovering and solving disagreement among investigators and between investigators and message receivers as such necessitate the validation of stimuli in experiment design. From the quantitative questions, the video *Empty Plate* was the top ranked video in terms of narrative persuasion. However, we also dropped this video after thorough comparison and analyses of perceived main message provided by the open-ended questions between researchers and target audience. The quantitative and qualitative responses revealed important differences in audience perceptions. Therefore, the mixed-method approach to message evaluation incorporating qualitative work and iteration demonstrates the risk of relying on traditional quantitative message evaluation strategies. The coding results also demonstrated the features of the video messages from diverse dimensions. Future researchers may choose the features they want to test message effects.

This study has a few limitations. First, the snowball approach to identify stimuli may missed potential messages that meet our inclusion criteria and the most recently published messages, which may influence the duplication for future studies. Second, the participants are college students who have relative higher education level than the general public, which may undermine the generalizability of the study. Third, the small sample size does not allow us to do more complex analyses to further understand the demographic matching. Finally, due to the strict inclusion criteria for this study, we ended up with the messages from only one health promotion campaign, which limited the future use of these messages.

4. Conclusion

Targeted marketing contributes to the overconsumption of sugary beverages, which contributes to obesity and diabetes disparities among African American and Latino populations in the U.S. Research is needed for a better understanding of how to manipulate message features to engender surface versus deep level tailoring, and individual versus social level appeals. Target audiences reported strong perception of persuasion to the video messages we identified. The final stimuli showed equivalent perceptions about message features between researchers and audiences. Our study theorized a typology of message tailoring and used a mixed-methods approach to identify and validate a pool of real-world exemplar messages that can be used as the basis for future studies that will advance understanding of tailoring effects. Future studies could use more systematic way to identify messages of other features such as messages of different format (narrative or non-narrative), modality (video or print), or messages with clearer age range of target audience to provide a more comprehensive message library and approach to tailored message design that can be used for message testing studies.

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Table 1.

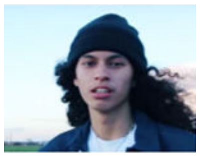
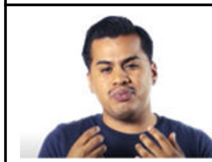
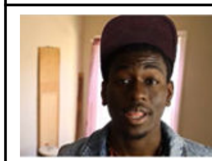
Descriptive results for ANOVA analyses for the overall narrative persuasiveness

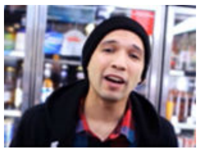
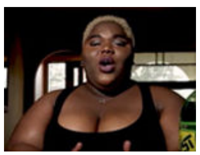
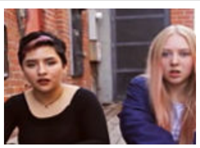
Comparison Category	Video Character Demographic		Participant Demographic		Overall Persuasiveness
ANOVA 1: Gender	Video with male characters	4.00 (0.65)	Male	3.87 (0.54)	P = 0.992 (Interaction effects between gender of participants and video characters)
			Female	4.04 (0.68)	
	Video with female characters	3.72 (0.70)	Male	3.60 (0.67)	
			Female	3.77 (0.72)	
	P = 0.019 (Main effects: gender of video characters)				
			Male total	3.72 (0.63)	P = 0.129 (Main effects of participant gender)
			Female total	3.90 (071)	
ANOVA 2: Race/ ethnicity	Video with Latino characters	3.89 (0.72)	Latino	3.91 (0.71)	P = 0.873 (Interaction effects between race/ethnicity of participants and video characters)
			Non-Latino	3.83 (0.74)	
	Video with African American characters	3.77 (0.60)	Latino	3.80 (0.59)	
			Non-Latino	3.67 (0.66)	
	P = 0.334 (Main effects: race/ ethnicity of video characters)				
			Latino total	3.88 (0.68)	P = 0.480 (Main effects of participant race/ethnicity)
			Non-Latino total	3.81 (0.72)	

Notes: Bolded text for statistical significance.

Table 2.

Message pool

Video	Surface Level Tailoring		Deep Structural Tailoring (values-based message appeals)		Level of Appeal in Ecological Framework		Overall Narrative Persuasion
	Character ethnicity identified by researcher % of participant identified character ethnicity	Character ethnicity identified by gender % of participant identified character gender	Familism identified by researcher % of participants identified Familism	Social justice identified by researcher % of participants identified social justice	Individual behavior change identified by researcher % of participants identified individual behavior change	SDOH identified by researcher % of participant identified SDOH	Mean SD
 Empty Plate (N=25)	Latino 100%	Male 100%	No 96%	Yes 80%	No 12%	Yes 92%	4.12 0.69
 A Taste of Home (N=22)	Latino 100%	Female 100%	Yes 100%	Yes 64%	No 36%	Yes 73%	4.09 0.68
 Lost in Translation (N=22)	Latino 100%	Male 100%	Yes 82%	Yes 68%	No 18%	Yes 73%	4.09 0.68
 Hearing No (N=22)	African American 100%	Male 100%	Yes 95%	No 0%	Yes 95%	Yes (Changed to No after validation) 0%	4.02 0.57

Video	Surface Level Tailoring		Deep Structural Tailoring (values-based message appeals)		Level of Appeal in Ecological Framework		Overall Narrative Persuasion
	Character ethnicity identified by researcher % of participant identified character ethnicity	**Character ethnicity identified by gender % of participant identified character gender**	**Familism identified by researcher % of participants identified Familism**	**Social justice identified by researcher % of participants identified social justice**	**Individual behavior change identified by researcher % of participants identified individual behavior change**	**SDOH identified by researcher % of participant identified SDOH**	**Mean SD**
 Bottled Up (N=27)	Latino 44%	Female 100%	No 52%	Yes 56%	No 41%	Yes 67%	3.79 0.62
 The Corner (N=24)	Latino 65%	Male 100%	No 95%	Yes 63%	Yes 50%	Yes 63%	3.76 0.64
 The Longest Mile (N=23)	African American 100%	Female 100%	Yes 91%	Yes 72%	Yes 65%	Yes 78%	3.53 0.53
 Monster (n=23)	Latino and White 70%	Female 100%	No 96%	No 35%	Yes 96%	Yes 30%	3.48 0.83